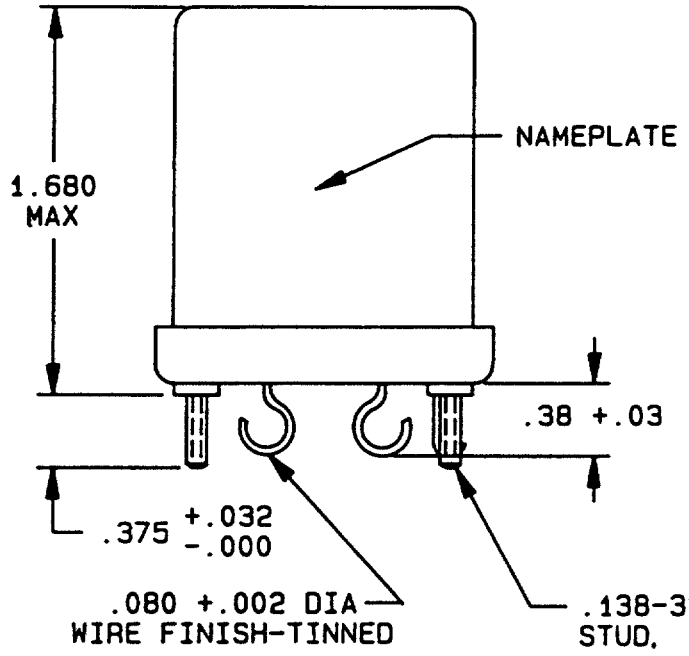
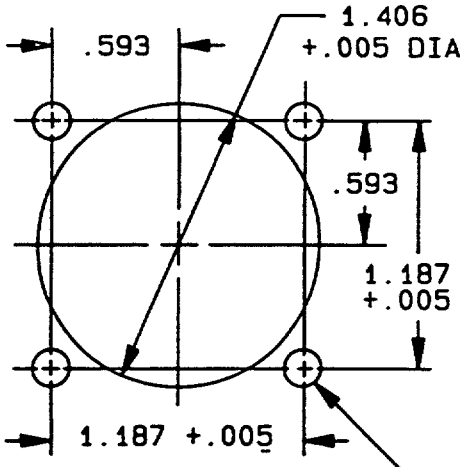
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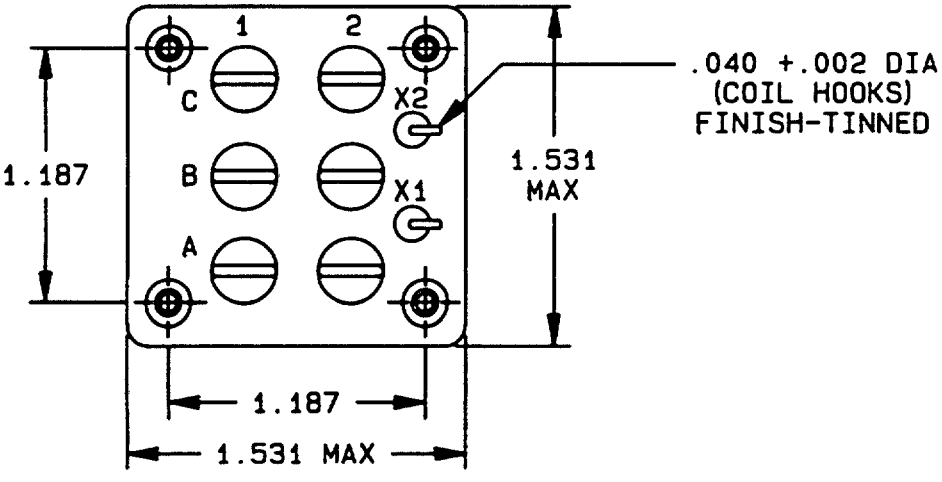
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SUGGESTED MOUNTING HOLE LAYOUT FOR STUD MOUNT



STUD MOUNT FOR -1A, -1B, -1C, AND -1D

(J) denotes changes

PREPARING ACTIVITY: Air Force-85 CUSTODIANS: ARMY - AIR FORCE - 85, NAVY - AS, DLA - ES REVIEW: Air Force- 11, 99 USER: PROJECT NUMBER: 5945-0836	MILITARY SPECIFICATION SHEET TITLE RELAYS, ELECTROMAGNETIC, 25 AMPERES, 3PST-NO, TYPE I, HERMETICALLY SEALED, HOOK AND SCREW TERMINALS, STUD AND BRACKET MOUNTED	SPECIFICATION SHEET NUMBER MS27418J 16 March 90 SUPERSEDING MS27418H 28 April 1989 AMSC N/A FSC 5945

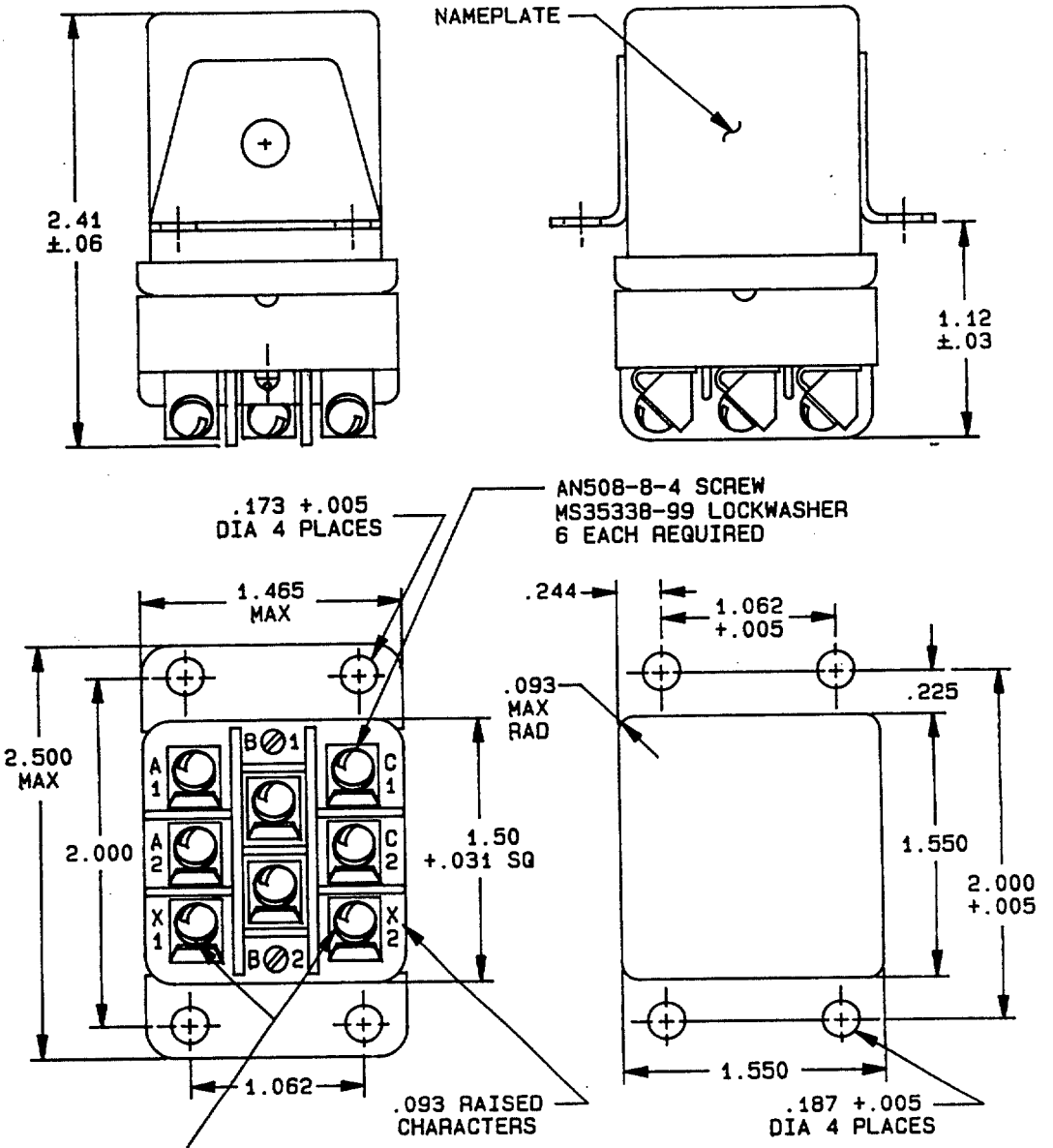
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BRACKET MOUNT FOR
-2A, -2B, -2C, AND -2D

SUGGESTED MOUNTING HOLE
LAYOUT FOR BRACKET MOUNT

PREPARING ACTIVITY: Air Force-85 CUSTODIANS: ARMY - NAVY - AS AIR FORCE - 85 DLA - ES REVIEW: Air Force- 11, 99 USER: PROJECT NUMBER: 5945-0836	MILITARY SPECIFICATION SHEET TITLE RELAYS, ELECTROMAGNETIC, 25 AMPERES, 3PST-NO, TYPE I, HERMETICALLY SEALED, HOOK AND SCREW TERMINALS, STUD AND BRACKET MOUNTED	SPECIFICATION SHEET NUMBER **MS27418J** SUPERSEDING MS27418H 28 April 1989 AMSC N/A FSC 5945

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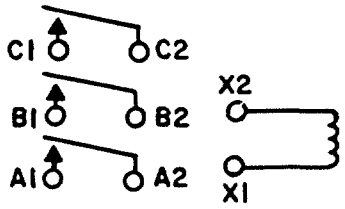
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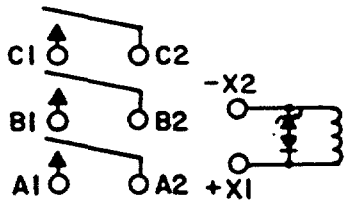
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CIRCUIT DIAGRAM A

(DC UNSUPPRESSED AND AC COILS)



CIRCUIT DIAGRAM B

(DC SUPPRESSED COILS)

Inches	mm	Inches	mm
.001	0.03	.244	6.20
.002	0.05	.375	9.52
.005	0.13	.38	9.7
.03	0.8	.593	15.06
.031	0.79	1.062	26.97
.032	0.81	1.12	28.5
.040	1.02	1.187	30.15
.06	1.5	1.406	35.71
.080	2.03	1.465	37.21
.093	2.36	1.50	38.1
.138	3.51	1.531	38.89
.152	3.86	1.550	39.37
.173	4.39	1.680	42.67
.187	4.75	2.000	50.80
.225	5.72	2.41	61.2
		2.500	63.50

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. A, B, and C contacts capable of accepting MS25036-53 lugs, X1 and X2 contacts capable of accepting MS25036-7 or -2 lugs.
5. Terminal numbers need not appear on relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
6. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.
7. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification sheet to the extent specified herein.

PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - NAVY - AS
AIR FORCE - DLA - ES
REVIEW: Air Force- 11, 99
USER:
PROJECT NUMBER: 5945-0836

MILITARY SPECIFICATION SHEET
TITLE RELAYS, ELECTROMAGNETIC, 25 AMPERES, 3PST-NO, TYPE I, HERMETICALLY SEALED, HOOK AND SCREW TERMINALS, STUD AND BRACKET MOUNTED

SPECIFICATION SHEET NUMBER
MS27418J
SUPERSEDING
MS27418H 28 April 1983
AMSC N/A FSC 5945

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TABLE I. Dash numbers and characteristics.

MS part number	Type	Coil	Circuit	Terminal type	Mounting or Mating socket	Max weight in pounds
MS27418-1A	I	ac	A	Solder hook	Stud	.50
MS27418-1B	I	dc	A	Solder hook	Stud	.50
MS27418-1C	I	ac	A	Solder hook	Stud	.50
MS27418-2A	I	ac	A	Screw	Bracket	.60
MS27418-2B	I	dc	A	Screw	Bracket	.60
MS27418-2C	I	ac	A	Screw	Bracket	.60
MS27418-1D	I	dc	B 1/ 2/	Solder hook	Stud	.51
MS27418-2D	I	dc	B 1/ 2/	Screw	Bracket	.61

1/ Transient voltage (back emf) 42 V dc maximum.
2/ Diodes shall have a peak inverse voltage of 600 V dc minimum.

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TABLE II. Operating characteristics.

MS part no. MS27418-	Coil data												Time (milliseconds maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 3/	Drop out voltage 3/		Operate 4/	Release 5/	Contact bounce			
		Volts 1/	Freq. Hz	Res Ω ±10%	Volts	Amperes	Normal 3/	High temp test	Cont current test						Main		Aux	
															NO	NC	NO	NC
1A	X1, X2	115	400	N/A	122	.055	95	100	108	40	5.0		25	50	2			
2A	X1, X2	115	400	N/A	122	.055	95	100	108	40	5.0		25	50	2			
1B	X1, X2	28	dc	160	29 2/	.25	18	19.5	22.5	7.0	1.5		20	10	2			
2B	X1, X2	28	dc	160	29 2/	.25	18	19.5	22.5	7.0	1.5		20	10	2			
1C	X1, X2	115	50/60	N/A	122	.06	95	100	108	40	5.0		25	50	2			
2C	X1, X2	115	50/60	N/A	122	.06	95	100	108	40	5.0		25	50	2			
1D	X1, X2	28	dc	160	29 2/	.25	18	19.5	22.5	7.0	1.5		20	10	2			
2D	X1, X2	28	dc	160	29 2/	.25	18	19.5	22.5	7.0	1.5		20	10	2			

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Maximum coil voltage shall be +32 V dc when maximum ambient temperature does not exceed 85°C.
3/ Over the temperature range.
4/ With nominal coil voltage.
5/ From nominal coil voltage.

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PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY- AS NAVY- AS AIR FORCE- 85 DLA- ES
REVIEW: Air Force- 11, 99
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PROJECT NUMBER: 5945-0836
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MILITARY SPECIFICATION SHEET
TITLE RELAYS, ELECTROMAGNETIC, 25 AMPERES, 3PST-NO, TYPE I, HERMETICALLY SEALED, HOOK AND SCREW TERMINALS, STUD AND BRACKET MOUNTED

SPECIFICATION SHEET NUMBER
MS27418J
SUPERSEDING MS27418H
28 April 1989
AMSC N/A
FSC 5945

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TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, I phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	50/60 Hz	400 Hz	60 Hz	400 Hz	50/60 Hz	400 Hz	60 Hz	
Resistive	50	25				25	25			25	25			
Inductive	10					25	25			25	25			2/
Inductive	10	15												
Motor	50	20				20	12			20	12			
Lamp	50	10				10	10			10	10			
Transfer load														3/
Mechanical life reduced current	200	6.3				6.3	6.3			6.3	6.3			
Intermediate current	Applicable in accordance with specification													

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ 0.7 pF, inductive.
3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - AS NAVY - AIR FORCE - 85 DLA - ES
REVIEW: Air Force- 11, 99
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Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 g's
Duration	11 ms
Max duration contact opening	2 ms
Vibration - sinusoidal	
Operating	
G-rating	10 g's
Frequency range	5 to 1,000 Hz
Nonoperation	
G-level	15 g's
Frequency range	20 to 2,000 Hz
Acceleration	10 g's

Electrical characteristics

Insulation resistance, initial		100 megohms
After life or environmental test		50 megohms
Dielectric strength (sea level)		
	Initial	After life tests
Coil to case	1,250 V rms	1,000 V rms
Aux contacts		
All other points	1,500 V rms	1,125 V rms
Dielectric strength (altitude)		80,000 ft
Coil to case		500 V rms
Aux contacts		
All other points		500 V rms
Max contact drop initial		0.150 volt
After life test		0.175 volt
Overload current		80 amperes dc 120 amperes ac
Rupture current		100 amperes dc 150 amperes ac
Duty rating		Continuous
RFI specification		MIL-STD-461

(Applicable to coil circuits of ac operated relays)

PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - NAVY - AS, AIR FORCE - 85, DLA - ES
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